

Lab 7 – Time Series Visualization with Bokeh

Dataset: `daily-minimum-temperatures-in-melbourne.csv`

```
In [2]: !pip install bokeh --quiet
import pandas as pd
import numpy as np
from bokeh.plotting import figure, show
from bokeh.io import output_notebook
from bokeh.models import (
    ColumnDataSource, HoverTool, DatetimeTickFormatter,
    NumeralTickFormatter, DateRangeSlider, CustomJS
)
from bokeh.layouts import row, column
from bokeh.transform import factor_cmap
from bokeh.palettes import Viridis10

output_notebook()

# — Chargement & nettoyage —————
df = pd.read_csv('daily-minimum-temperatures-in-melbourne.csv')
df.columns = ['Date', 'Temperature']
df['Date'] = pd.to_datetime(df['Date'])
df['Temperature'] = df['Temperature'].astype(str).str.replace('?', '', regex=False)
df['Temperature'] = pd.to_numeric(df['Temperature'], errors='coerce')
df = df.dropna()
print(df.shape)
df.head()
```

(3650, 2)

```
Out[2]:
```

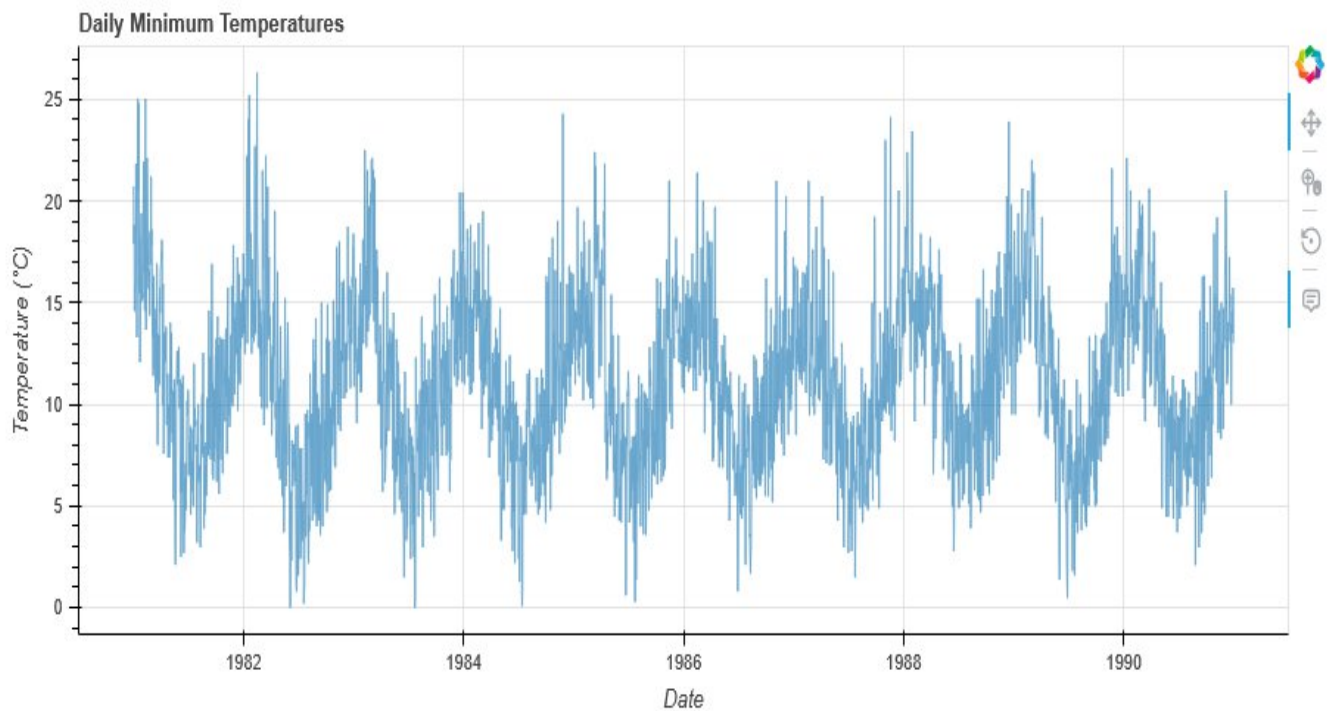
	Date	Temperature
0	1981-01-01	20.7
1	1981-01-02	17.9
2	1981-01-03	18.8
3	1981-01-04	14.6
4	1981-01-05	15.8

Question 1 – Basic Time Series Line Plot

```
In [3]: source1 = ColumnDataSource(df)

hover1 = HoverTool(tooltips=[
    ('Date', '@Date{%F}'),
    ('Temperature', '@Temperature{0.0} °C')
], formatters={'@Date': 'datetime'})

p1 = figure(
    title='Daily Minimum Temperatures',
    x_axis_type='datetime',
    x_axis_label='Date',
    y_axis_label='Temperature (°C)',
    width=900, height=380,
    tools=[hover1, 'pan', 'wheel_zoom', 'reset'],
)
p1.line('Date', 'Temperature', source=source1,
        line_color='#2980b9', line_width=1, alpha=0.7)
p1.xaxis.formatter = DatetimeTickFormatter(years='%Y', months='%b %Y')
show(p1)
```



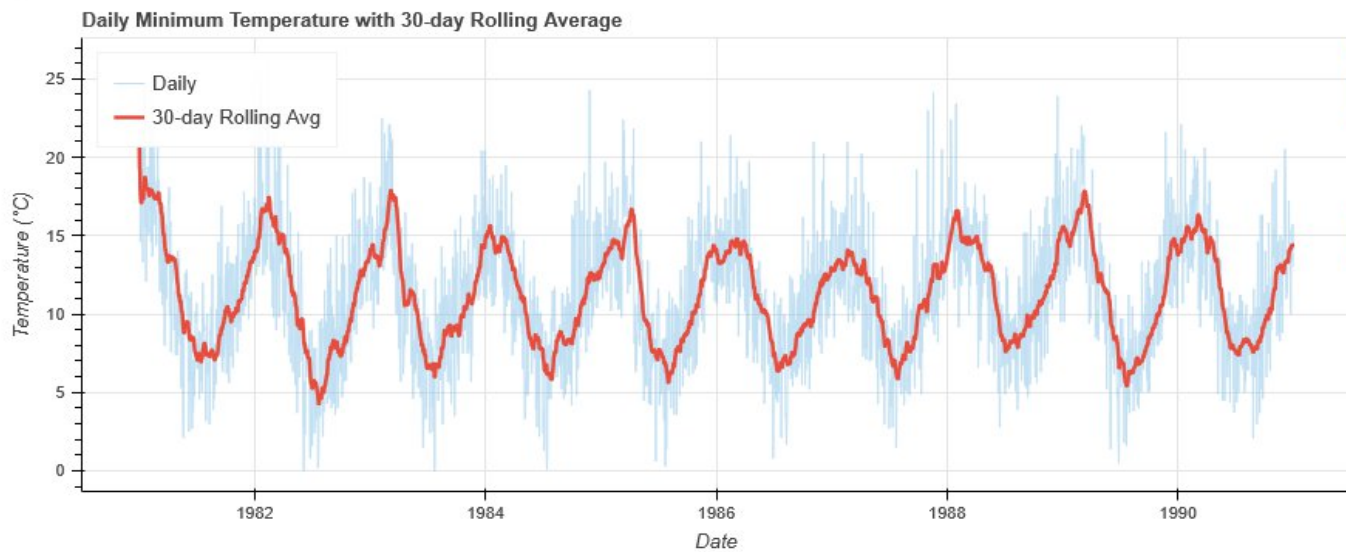
Question 2 – 30-day Rolling Average

```
In [4]: df['Rolling_Avg'] = df['Temperature'].rolling(30, min_periods=1).mean()
source2 = ColumnDataSource(df)

hover2 = HoverTool(tooltips=[
    ('Date', '@Date{%F}'),
    ('Temperature', '@Temperature{0.0} °C'),
    ('Rolling Avg', '@Rolling_Avg{0.0} °C'),
], formatters={'@Date': 'datetime'})

p2 = figure(
    title='Daily Minimum Temperature with 30-day Rolling Average',
    x_axis_type='datetime',
    x_axis_label='Date', y_axis_label='Temperature (°C)',
    width=900, height=380,
    tools=[hover2, 'pan', 'wheel_zoom', 'reset'],
)
p2.line('Date', 'Temperature', source=source2,
        line_color='#85c1e9', line_width=1, alpha=0.5, legend_label='Daily')
p2.line('Date', 'Rolling_Avg', source=source2,
        line_color='#e74c3c', line_width=2.5, line_dash='solid',
        legend_label='30-day Rolling Avg')
p2.legend.location = 'top_left'
p2.xaxis.formatter = DatetimeTickFormatter(years='%Y', months='%b %Y')
show(p2)
```

```
show(p2)
```



Question 3 – Monthly Box Plots

```
In [5]: month_names = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
                        'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
df['Month'] = df['Date'].dt.month
df['Month_name'] = df['Date'].dt.month.apply(lambda m: month_names[m-1])

# Statistiques par mois
grp = df.groupby('Month')['Temperature']
stats3 = pd.DataFrame({
    'month': month_names,
    'q1': grp.quantile(0.25).values,
    'q2': grp.quantile(0.50).values,
    'q3': grp.quantile(0.75).values,
    'upper': grp.quantile(0.95).values,
    'lower': grp.quantile(0.05).values,
})

src3 = ColumnDataSource(stats3)

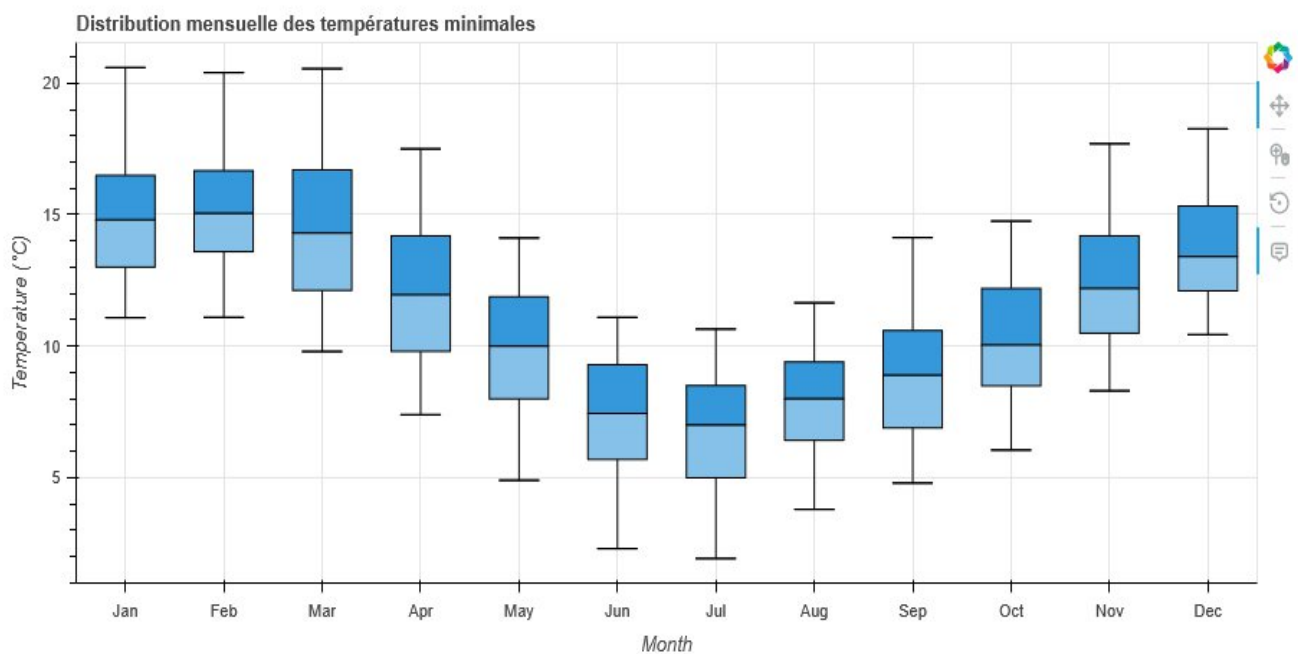
hover3 = HoverTool(tooltips=[
    ('Month', '@month'),
    ('Median', '@q2{0.1} °C'),
    ('Q1', '@q1{0.1} °C'),
    ('Q3', '@q3{0.1} °C'),
    ('Min 5%', '@lower{0.1} °C'),
    ('Max 95%', '@upper{0.1} °C'),
])

p3 = figure(
    title='Distribution mensuelle des températures minimales',
    x_range=month_names,
    x_axis_label='Month', y_axis_label='Temperature (°C)',
    width=900, height=420,
    tools=[hover3, 'pan', 'wheel_zoom', 'reset'],
)
```

```

p3 = figure(
    title='Distribution mensuelle des températures minimales',
    x_range=month_names,
    x_axis_label='Month', y_axis_label='Temperature (°C)',
    width=900, height=420,
    tools=[hover3, 'pan', 'wheel_zoom', 'reset'],
)
# Moustaches
p3.segment('month', 'upper', 'month', 'q3', source=src3, line_color='black')
p3.segment('month', 'lower', 'month', 'q1', source=src3, line_color='black')
# Boîtes
p3.vbar('month', 0.6, 'q2', 'q3', source=src3, fill_color='#3498db', line_color='black')
p3.vbar('month', 0.6, 'q1', 'q2', source=src3, fill_color='#85c1e9', line_color='black')
# Caps
p3.rect('month', 'upper', 0.4, 0.01, source=src3, line_color='black')
p3.rect('month', 'lower', 0.4, 0.01, source=src3, line_color='black')
show(p3)

```



Question 4 – Yearly Box Plots with factor_cmap

```

[6]: df['Year'] = df['Date'].dt.year.astype(str)
     years = sorted(df['Year'].unique())

     grp4 = df.groupby('Year')['Temperature']
     stats4 = pd.DataFrame({
         'year': years,
         'q1': [grp4.get_group(y).quantile(0.25) for y in years],
         'q2': [grp4.get_group(y).quantile(0.50) for y in years],
         'q3': [grp4.get_group(y).quantile(0.75) for y in years],
         'upper': [grp4.get_group(y).quantile(0.95) for y in years],
         'lower': [grp4.get_group(y).quantile(0.05) for y in years],
     })
     src4 = ColumnDataSource(stats4)

     palette = Viridis10[:len(years)]
     cmap = factor_cmap('year', palette=palette, factors=years)

```

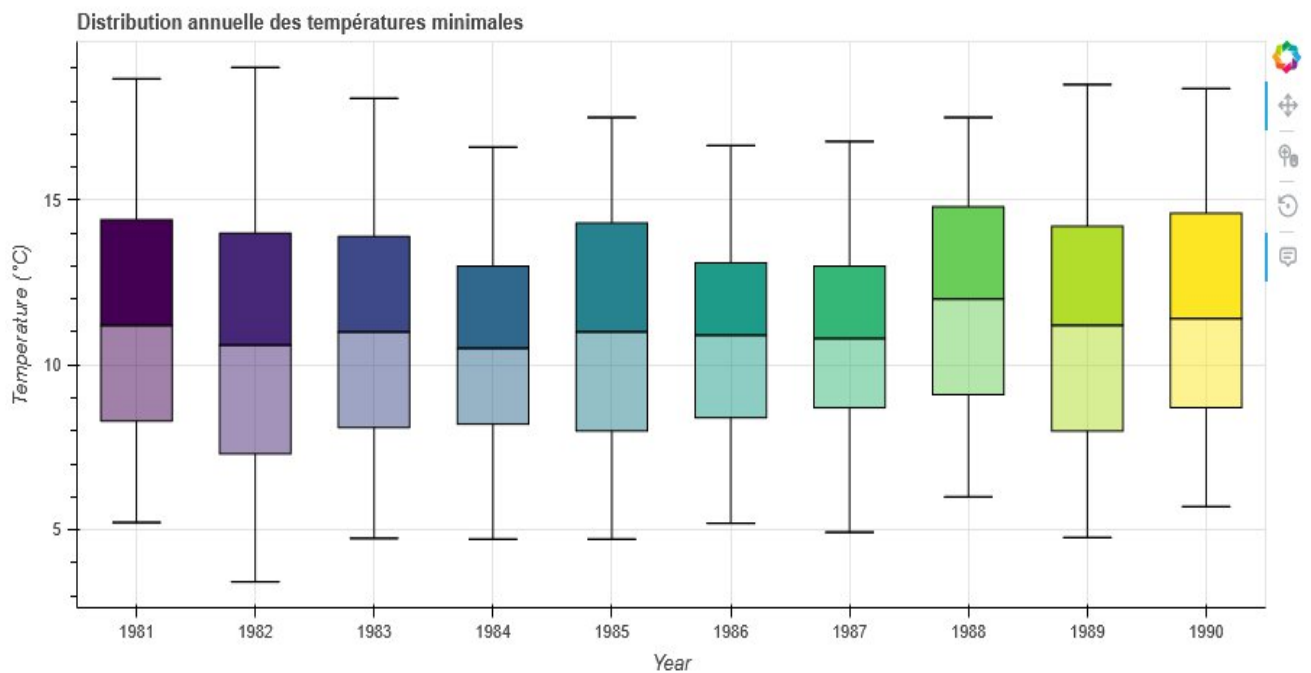
```

hover4 = HoverTool(tooltips=[
    ('Year', '@year'),
    ('Median', '@q2{0.1} °C'),
    ('Q1', '@q1{0.1} °C'),
    ('Q3', '@q3{0.1} °C'),
    ('Min 5%', '@lower{0.1} °C'),
    ('Max 95%', '@upper{0.1} °C'),
])

p4 = figure(
    title='Distribution annuelle des températures minimales',
    x_range=years,
    x_axis_label='Year', y_axis_label='Temperature (°C)',
    width=900, height=420,
    tools=[hover4, 'pan', 'wheel_zoom', 'reset'],
)

p4.segment('year', 'upper', 'year', 'q3', source=src4, line_color='black')
p4.segment('year', 'lower', 'year', 'q1', source=src4, line_color='black')
p4.vbar('year', 0.6, 'q2', 'q3', source=src4, fill_color=cmap, line_color='black')
p4.vbar('year', 0.6, 'q1', 'q2', source=src4, fill_color=cmap, line_color='black', fill_alpha=0.5)
p4.rect('year', 'upper', 0.4, 0.01, source=src4, line_color='black')
p4.rect('year', 'lower', 0.4, 0.01, source=src4, line_color='black')
show(p4)

```



Question 5 – Interactive Time Range Selection (DateRangeSlider)

```
In [7]: from bokeh.models import DateRangeSlider, CustomJS

src5_full = ColumnDataSource(df)
src5_filtered = ColumnDataSource(df.copy())

hover5 = HoverTool(tooltips=[
    ('Date', '@Date{%F}'),
    ('Temperature', '@Temperature{0.0} °C')
], formatters={'@Date': 'datetime'})

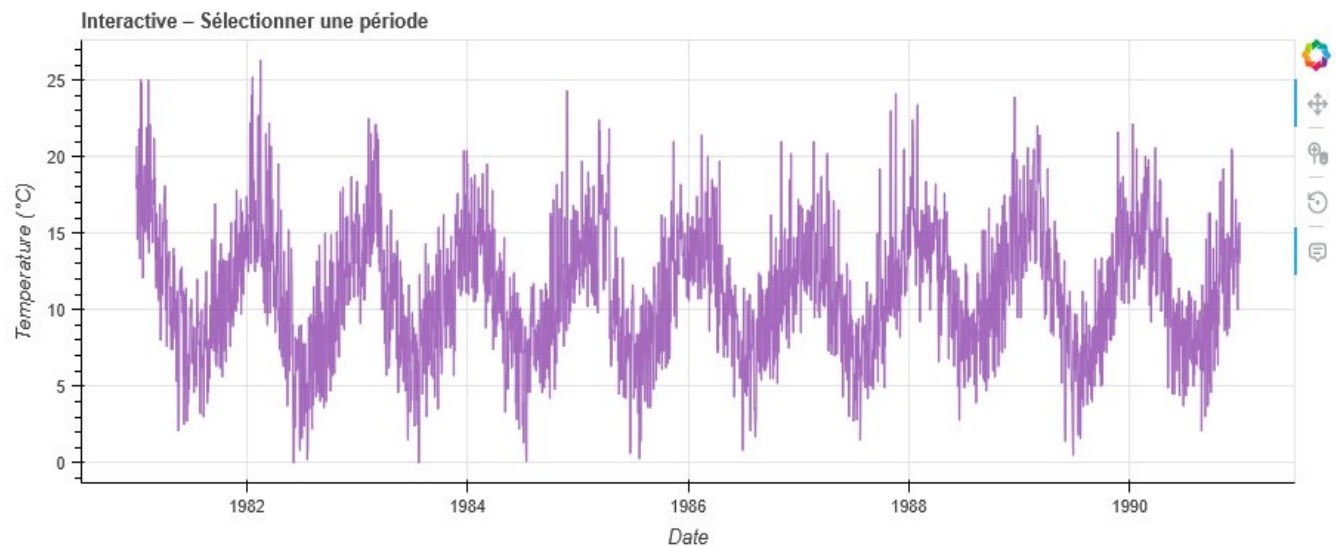
p5 = figure(
    title='Interactive - Sélectionner une période',
    x_axis_type='datetime',
    x_axis_label='Date', y_axis_label='Temperature (°C)',
    width=900, height=350,
    tools=[hover5, 'pan', 'wheel_zoom', 'reset'],
)
p5.line('Date', 'Temperature', source=src5_filtered,
        line_color='#8e44ad', line_width=1.2, alpha=0.8)
p5.xaxis.formatter = DatetimeTickFormatter(years='%Y', months='%b %Y')

slider5 = DateRangeSlider(
    title='Période',
    value=(df['Date'].min().timestamp()*1000,
          df['Date'].max().timestamp()*1000),
    start=df['Date'].min().timestamp()*1000,
    end=df['Date'].max().timestamp()*1000,
    width=880
)

callback5 = CustomJS(args=dict(src_full=src5_full, src_filt=src5_filtered, slider=slider5), code="""
    const [start, end] = slider.value;
    const dates = src_full.data['Date'];
    const temps = src_full.data['Temperature'];
    const new_dates = [];
    const new_temps = [];
    for (let i = 0; i < dates.length; i++) {
        if (dates[i] >= start && dates[i] <= end) {
            new_dates.push(dates[i]);
            new_temps.push(temps[i]);
        }
    }
    src_filt.data = {Date: new_dates, Temperature: new_temps};
    src_filt.change.emit();
""")
slider5.js_on_change('value', callback5)

show(column(slider5, p5))
```

Période: 01 Jan 1981 .. 31 Dec 1990



Question 6 – Time Series Decomposition (trend + saisonnalité)

```
[8]: # — Rééchantillonnage mensuel —————
df_month = df.set_index('Date').resample('ME')['Temperature'].mean().reset_index()
df_month.columns = ['Date', 'Temperature']

# Tendence : moyenne mobile 12 mois
df_month['Trend'] = df_month['Temperature'].rolling(12, center=True, min_periods=1).mean()

# Saisonnalité = signal original - tendance
df_month['Seasonal'] = df_month['Temperature'] - df_month['Trend']

src_orig = ColumnDataSource(df_month)
src_trend = ColumnDataSource(df_month.dropna(subset=['Trend']))
src_seas = ColumnDataSource(df_month.dropna(subset=['Seasonal']))

def make_hover(val_label, val_col):
    return HoverTool(tooltips=[
        ('Date', '@Date{%b %Y}'),
        (val_label, f'@{val_col}{{0.00}} °C'),
    ], formatters={'@Date': 'datetime'})

fmt = DatetimeTickFormatter(years='%Y', months='%b %Y')

# Graphe 1 : données originales
pa = figure(title='Données originales (moyenne mensuelle)',
            x_axis_type='datetime', x_axis_label='Date',
            y_axis_label='Temp (°C)', width=900, height=250,
            tools=[make_hover('Temperature', 'Temperature'), 'pan', 'wheel_zoom', 'reset'])
pa.line('Date', 'Temperature', source=src_orig, line_color='#2980b9', line_width=2)
pa.xaxis.formatter = fmt
```

```
# Graphe 2 : tendance
pb = figure(title='Tendance (rolling 12 mois)',
            x_axis_type='datetime', x_axis_label='Date',
            y_axis_label='Temp (°C)', width=900, height=250,
            x_range=pa.x_range,
            tools=[make_hover('Trend', 'Trend'), 'pan', 'wheel_zoom', 'reset'])
pb.line('Date', 'Trend', source=src_trend, line_color='#e74c3c', line_width=2)
pb.xaxis.formatter = fmt

# Graphe 3 : saisonnalité
pc = figure(title='Composante saisonnière',
            x_axis_type='datetime', x_axis_label='Date',
            y_axis_label='Résidu (°C)', width=900, height=250,
            x_range=pa.x_range,
            tools=[make_hover('Seasonal', 'Seasonal'), 'pan', 'wheel_zoom', 'reset'])
pc.line('Date', 'Seasonal', source=src_seas, line_color='#27ae60', line_width=2)
pc.xaxis.formatter = fmt

show(column(pa, pb, pc))
```

